

EXECUTIVE BRIEFING DOCUMENT

EDTECH STUDENT RETENTION & SUBSCRIPTION CHURN DIAGNOSTIC BRIEFING

TO: Head of Student Success & Chief Product Officer (CPO)
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SUBJECT: Diagnostics & Action Plan for 14-Day Early Student Withdrawals and Assessment Bottlenecks

1. EXECUTIVE SUMMARY

Net subscriber lifetime value (LTV) is severely bottlenecked by early course withdrawals. Diagnostic analysis of 32,000+ student registrations and 10M+ VLE clickstream interactions reveals that 68% of all student course withdrawals occur within the first 14 days of enrolment.

Statistical testing confirmed significant differences in early study time between churned and retained learners. Implementing automated Day-5 study intervention nudges for low-activity learners and restructuring high-friction assessment modules is projected to reduce early student churn by 8–12%.

2. KEY ANALYTICAL FINDINGS

- **The Week 2 Risk Threshold (2x Churn Spike):**
Students logging under 15 minutes of active platform time in Week 2 demonstrate a 30.5% overall course withdrawal rate—nearly DOUBLE (1.91x) the churn rate of optimal learners (16.0% churn at 180+ active minutes).
- **Active Engagement Decay:**
Across all presentation cohorts, active learning decay accelerates rapidly between Week 1 and Week 4, dropping by an average of ~52% prior to the first major assessment deadline.
- **Assessment Failure Bottlenecks:**
High-friction Tutor Marked Assessments (TMAs) in specific foundational modules (e.g., BBB and DDD) exhibit failure rates spiking up to 48%, acting as the primary catalyst for log-off and eventual subscription cancellation.

3. STRATEGIC RECOMMENDATIONS & ACTION PLAN

1. **Triggered Early Interventions (Day 5 Nudges):**
Deploy automated, dynamic study reminders and tutor outreach on Day 5 when a student's accumulated Week 1–2 study time drops below 45 minutes.
2. **Modularise High-Friction Assessments:**
Restructure high-failure TMAs into bite-sized micro-quizzes to flatten the difficulty curve and prevent early course drop-offs.
3. **Financial Impact:**
Targeting the 14-day early drop-off window is estimated to recover 8–12% of churned monthly subscription revenue.

4. TECHNICAL METHODOLOGY & DATA ENGINE

- **Data Engine:** DuckDB (SQL) for high-performance extraction and staging of 10M+ clickstream logs.
- **Statistical Analytics:** Python (Pandas, SciPy) for Welch's t-test hypothesis testing and cohort decay.
- **Business Intelligence:** Tableau Public 2-Page Executive Dashboard.